

On the Sensitivity of COCOMO II Software Cost Estimation Model

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Abstract

Software cost estimation techniques predict the amount of effort required to develop a software system. Cost estimates are needed throughout the software lifecycle to determine feasibility of software projects and to provide for appropriate allocation or reallocation of available resources. As the cost of a project depends on the nature and characteristics of the project, the accuracy of the estimates depends on the amount of reliable information about the product to be developed. At the same time, most cost estimation models rely heavily on subjective expert evaluations affected by possibly high degree of imprecision and uncertainty. To assess the effect of imprecise evaluations, a comprehensive sensitivity analysis has been performed on a major cost estimation model, COCOMO II. Results of this analysis are described and explicated in this paper. To reduce risk of drawing biased conclusions, three different methods for sensitivity analysis have been employed: mathematical analysis of the estimating equation, Monte Carlo simulation, and error propagation. The results of the first two methods are very consistent and confirm expected highest sensitivity of the model to the imprecision of the size estimate. Error propagation allows determination of the combined impact of imprecision in multiple inputs and it is therefore most valuable from the practical point of view. The results obtained by this technique also indicate very strong sensitivity to the imprecision in size estimates. A possible way to cope with imprecise information in software cost estimation is indicated in the concluding part of the paper.

1. Introduction

Reliable and accurate estimates of software development cost and effort are needed

throughout the software development cycle to determine feasibility of software projects and to provide for appropriate allocation or reallocation of available resources. This means that the task of software cost estimation needs to be performed very early in the development cycle.

The cost of a software project depends on the nature and characteristics of the project and, therefore, the accuracy of the estimates depends on the amount of reliable information about the product to be developed. Obviously the amount and quality of information available at the very beginning of any project is limited. However, most cost estimation models rely heavily on such data obtained typically by subjective expert evaluations affected by high degree of imprecision and uncertainty. This is also the case for one of the newest additions to the family of cost estimation models, the recently revised COCOMO II. For this reason, this model has been chosen as a subject of the analytical and experimental treatment described here. The model is briefly described in Section 2 along with a concise introduction of software cost estimation models in general. To assess the effect of imprecise evaluations, a comprehensive sensitivity analysis has been performed on the selected model. Results of the sensitivity study are described in great detail in Section 3. It describes three complementary approaches to assess the sensitivity of mathematical models. Together, the results obtained using these three techniques provide a complete picture of the sensitivity of the model to the variation of its independent variables. Qualitatively, the results follow our intuition: the most critical independent variable is the projected size of a software system. Quantitatively, the results are quite alarming, indicating that sensitivity to the imprecision in size estimates is very high. A possible way to cope with imprecise information in software cost estimation is indicated in the concluding part of the paper, Section 4.

2. Software Cost Estimation

Software cost estimation is of great interest to software project managers who need to predict project costs during early phases in the software life cycle. As a result, many different models of software cost and effort estimation have been developed and used during the past four decades. They often share a common approach: effort is expressed as a predefined function of one or more variables (such as the size of the product, experience of the development team, level of reuse, etc.).

2.1. Cost Estimation Techniques

Since 1960's, great research efforts have been devoted to development of software cost estimation methodologies and techniques [2, 5, 7-14]. Today, there are many academic and commercial tools implementing some of these techniques. Most of practically used tools are based on models introduced by Albrecht [3, 4], Boehm [2, 14], and Putnam [5].

- The FPA (Function Point Analysis) model introduced by Albrecht [3] estimates the size of software projects using functional metrics and determines the cost/effort as a product of function points required for the new system and productivity of the development team/process.
- CoCoMo (Constructive Cost Model) was introduced by Boehm [2] and underwent several major revisions [8, 13, 14]. The model uses direct estimates of the project size and large number of coefficients called effort multipliers and scale factors.
- SLIM (Software Lifecycle Model) was introduced by Putnam [5]. Compared to CoCoMo, it exhibits very high rate of scaling with the project size that is also estimated directly. The number of parameters is much lower than that of COCOMO.

In the following section, the COCOMO II is described in more details. The reasons for choosing this particular model are twofold. First, it represents the newest developments in the field of software cost estimation. Second, it uses large number of parameters and is, therefore, suitable for sensitivity study. In addition, the authors of this model have provided us with a sample dataset that has made this study possible.

2.2. COCOMOII Model

The cost estimation model COCOMO II [14] has been developed as a major revision of the original COCOMO model [2] by group of industrial and academic affiliates at the University of Southern California. This group reviewed the revised set of COCOMO equations and has been collecting data to evaluate hypotheses to select the significant parameters for COCOMOII. The new model explicitly handles the availability of additional information in later stages of a project, the nonlinear costs of reusing software components, and the effects of several factors on the diseconomies of scale (such as the turnover rate of the staff, the geographic dispersion of the team, and the maturity of the development process). It also revises some coefficient values and eliminates discontinuities present in the old model (related to development mode and maintenance vs. adaptation).

COCOMOII consists of three submodels, each one offering increased fidelity. Listed in increasing fidelity, these submodels are called Applications Composition, Early Design, and Post-architecture models. Until recently, only the last and most detailed submodel, Post-architecture, had been implemented in a calibrated software tool. As such, unless otherwise explicitly indicated, all further references in this study to COCOMO II can be assumed to be the Post-architecture model [13].

COCOMOII model estimates the effort to develop a software system based on its projected size S , effort multipliers EM_i and exponent scale factors W_i

$$PM = 2.94 \prod_{i=1}^{17} EM_i S^{0.91+0.01 \sum_{i=1}^5 W_i}, \quad (1)$$

where PM is the effort expressed in person-months, S is the projected size of the software project (expressed in thousands of lines of code KLOC), EM_i ($i = 1, 2, \dots, 17$) are effort multipliers, and W_i ($i = 1, 2, \dots, 5$) are exponent scale factors.

There are 17 effort multipliers capturing the characteristics of the software development that affect the effort to complete the project. These weighted multipliers are grouped into four categories (product, platform, personnel and project) and their product is used to adjust the effort. The nominal weight assigned to each

multiplier is 1.0. If a rating level has a detrimental effect on effort, its corresponding multiplier is greater than 1.0. Conversely, if the rating level reduces the effort then the corresponding multiplier is less than 1.0.

There are five exponent scale factors (precedentness, development flexibility, architecture/risk resolution, team cohesion and process maturity). They account for the relative economies or diseconomies of scale encountered as a software project increases its size based on different nominal values and rating schemes.

3. Study of the Sensitivity

To examine sensitivity of the model (its dependent variable PM) to variations of the independent variables (S , EM_i and W_i), three groups of experiments/analytical evaluations have been performed: mathematical analysis, Monte Carlo simulation, and error propagation.

3.1. Mathematical analysis

In order to analyze the model sensitivity analytically, the derivatives of the dependent variable with respect to each independent variable are determined to quantitatively evaluate the sensitivities to different input variables. Subsequently, tolerable error values for EM_i , S and W_i are determined later to find out the most critical input variable.

The sensitivity of effort to EM_i can be determined using the following derivative equation

$$\frac{\partial PM}{\partial EM_i} = \prod_{\substack{j=1 \\ j \neq i}}^{17} EM_j \cdot 2.94 \cdot S^{0.91+0.01 \sum_{j=1}^5 W_j} \quad (2)$$

Based on these equations, the tolerable EM_i relative errors have been determined using available data on 15 projects. The threshold of the relative error of effort has been set to the value 5%. The results are summarized in Table 1.

Table 1. Sensitivity of effort E to effort multipliers EM

P	EM_1	EM_2	EM_3	EM_4	EM_5	EM_6	EM_7	EM_8	EM_9	EM_{10}	EM_{11}	EM_{12}	EM_{13}	EM_{14}	EM_{15}	EM_{16}	EM_{17}
1	5.35%	5.31%	5.35%	5.35%	5.35%	5.35%	5.86%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%
2	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%
3	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%
4	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%
5	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	4.83%	5.00%
6	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%
7	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%
8	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%
9	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%
10	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%
11	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%
12	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%	5.35%
13	4.35%	4.35%	4.01%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.35%	4.33%	4.35%
14	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%
15	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%	5.00%

Table 2. Sensitivity of effort E to size S

P.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
S	4.45%	4.43%	4.39%	4.85%	4.46%	4.46%	4.46%	3.80%	4.82%	3.75%	5.19%	5.23%	3.78%	4.32%	4.64%

Table 3. Sensitivity of effort E to exponent scale factors W_i

P.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
W_1	22.01%	49.71%	28.20%	42.14%	31.72%	21.43%	23.31%	48.40%	36.40%	21.51%	126.9%	105.4%	20.93%	25.05%	36.54%
W_2	21.53%	40.55%	27.58%	30.92%	38.84%	20.96%	22.81%	35.51%	26.71%	32.92%	155.8%	129.4%	32.04%	38.35%	74.52%
W_3	15.44%	19.38%	32.98%	55.39%	37.10%	25.06%	27.27%	31.87%	47.84%	23.60%	55.59%	46.17%	30.61%	27.49%	80.06%
W_4	24.92%	56.29%	42.51%	71.57%	47.82%	48.52%	52.79%	41.11%	61.83%	30.44%	N/A	N/A	39.45%	47.21%	N/A
W_5	13.99%	15.80%	25.61%	28.71%	33.62%	22.71%	24.71%	38.47%	21.70%	28.49%	22.41%	20.94%	20.80%	28.45%	48.41%

The tolerable error values range within 4.01% and 5.35% with the average of 4.99%. For the same project data, the values are very close to each other, and the values of a single input across all project data are also quite similar.

The sensitivity of effort to input variable S can be determined by the equation

$$\frac{\partial PM}{\partial S} = \quad (3)$$

and the sensitivity to exponential scale factors W_i by the equation

$$\frac{\partial PM}{\partial W_i} = \quad (4)$$

The results, assuming the threshold of the relative error of effort to be again 5%, are summarized in Tables 2 and 3.

Table 2 shows the tolerable relative errors of S that are quite similar across all 15 projects. They range from 3.75% to 5.23% with the average value of 4.46%. The results for W_i , shown in Table 3, have much wider range (13.99% - 155.80%) and the average value of 40.04%. Four notably large values are encountered for W_1 and W_2 for projects 11 and 12. This means that the preset threshold value of error of effort (5%) is reached only for very large errors ($> 100\%$) of the corresponding coefficients. This could be explained by very small values of these coefficients for given projects. The values of sensitivity to W_4 for projects 11, 12 and 15 are not available (N/A) because the actual values of W_4 in these three projects are zero, thus preventing proper calculations (division by zero).

As the results obtained the three types of input variables (EM_i , S and W_i) are quite uniform within each group, they have been grouped and averaged to find the order of impact of their inaccuracy on the error of the estimated effort. For convenience the results are summarized in the following table.

Table 4. Average sensitivity to three groups of parameters

Group Of Parameters	Average Tolerable Error
EM_i	4.99%
S	4.47%
W_i	40.04%

Evidently projected size S is the most critical parameter as the estimates are most sensitive to it. It is closely followed by effort multipliers EM_i whose impact is only slightly smaller. Finally, the exponent scale factors W_i exhibit relative low impact on the accuracy of the estimates. Implications of this finding are explained later in the paper taking in account results of other sensitivity analysis methods.

3.2. Monte Carlo Simulation

Monte Carlo simulation is applied to a single error input case, and it is assumed that only one input variable has error in one simulation run as shown in Figure 1. The distribution of the error was initially considered to be normal that is fully justifiable for size S that does not have any strict upper limit of range. Since normal distribution has no limited range, 99.74% of the data fall within the goal range by making the standard deviation small enough. This is based on the well-known fact that within the range of 6 times standard deviation, a normal distribution covers 99.74% of data [15]. This fact also allows to apply normal distribution to EM_i and W_i , although both of them do have strict

limited range. To make the experiments closer to actual conditions, two other distributions have been used to model EM_i and W_i : triangular and beta distribution that have limited ranges. These additional distributions have not been applied to S as it has no upper limit defined. Each simulation run was performed using a random sample of $N=1000$ values generated randomly with given distribution. The standard deviation was set to 0.001 in the first run and gradually increased by 0.001 up to 0.3. The boundary

value 0.3 was determined by examining the standard deviation to make the distributed data fall within valid range. The mean of the normal distribution was set to the actual values of S from the dataset. Mean relative error MRE has been determined for each simulation run. The relative error has been defined as the absolute difference of each simulation result PM and the mean value of all results in the run divided by this mean value. MRE is then the mean of all relative errors in a given run.

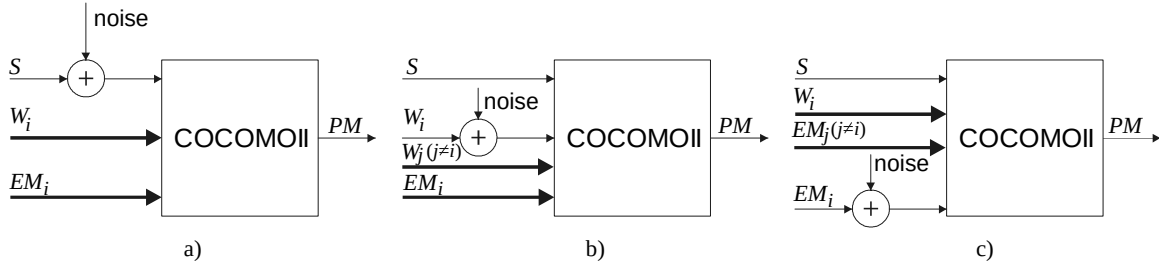


Figure 1. Monte Carlo simulation for single input error in: (a) size, (b) effort multipliers, (c) exponent scale factors

The mean relative errors are examined with the goal to find the largest input variable error corresponding to the mean relative error of the effort which is not greater than the threshold of 5% of the actual value of effort PM . This largest error, called tolerable input error, is a reflection of sensitivity. Tolerable input errors for given dataset and all three groups of parameters (EM_i , S and W_i) are shown in Table 5.

The table summarizes the tolerable input errors of S , W_i and EM_i found using three different distributions of input variables for all 15 project data. Similarly to the results obtained by mathematical analysis, values of tolerable input error for W_i in projects 11 and 12 were not determined which means that the 5% of the effort error is never obtained even when W_i has input error of 100% of the actual W_i .

Table 5. Tolerable input errors found using Monte Carlo simulation

Distribution Variable	Normal			Triangular		Beta	
	S	W_i	EM_i	W_i	EM_i	W_i	EM_i
Project 1	5.10%	25.80%	6.20%	24.10%	5.80%	32.70%	10.10%
Project 2	5.10%	49.70%	6.40%	47.60%	6.10%	60.30%	10.90%
Project 3	5.10%	25.80%	6.10%	24.10%	6.00%	32.70%	9.80%
Project 4	5.10%	33.10%	6.30%	34.60%	6.00%	39.50%	10.70%
Project 5	5.20%	25.80%	6.10%	29.10%	6.00%	32.70%	9.80%
Project 6	5.10%	25.80%	6.10%	24.50%	6.00%	32.70%	9.80%
Project 7	5.20%	25.80%	6.10%	24.20%	6.10%	32.70%	9.80%
Project 8	5.10%	33.10%	6.10%	34.50%	6.00%	39.50%	9.80%
Project 9	5.10%	33.10%	6.30%	34.60%	6.00%	39.50%	10.70%
Project 10	5.20%	20.90%	6.10%	46.00%	6.10%	27.80%	9.80%
Project 11	5.10%	N/A	6.20%	N/A	5.80%	N/A	10.10%
Project 12	5.10%	N/A	6.20%	N/A	5.80%	N/A	10.10%
Project 13	5.20%	20.90%	6.10%	45.50%	6.00%	27.80%	9.80%
Project 14	5.10%	20.90%	6.30%	46.00%	6.00%	27.80%	10.70%
Project 15	5.20%	20.90%	6.20%	25.40%	5.80%	27.80%	10.10%
Average	5.13%	27.82%	6.19%	33.86%	5.97%	38.37%	10.13%

The bottom row of the Table 5 shows average values of tolerable input error for all three groups of parameters and the three random distributions considered. Overall, the results obtained by Monte Carlo simulation correspond

very well to these obtained using mathematical analysis (see Table 4): the COCOMO II model is most sensitive to error in projected size S , followed by effort multipliers EM_i , with

exponent scale factors W_i being the least important.

3.3. Error Propagation

Error propagation is applied in order to determine the impact of the cumulative input noise on the effort estimation. Shown in Figure 2, error propagation is a way of combining two or more random errors together to get a new variable representing the cumulative error. It is used in situations where it is not possible to measure the same parameter several times and thereby estimate the random error on the final result directly. It can be used when there is a need to measure more than one quantity to get the final result.

There are four basic rules [16, 17] to determine error propagation, assuming that the errors are normal in nature. In general it is assumed that a condition cannot be observed directly, but depends on other quantities that can be measured, e.g. x and y . The absolute uncertainties (standard deviation of Gaussian distribution) on these measurements are $\pm\delta x$ and $\pm\delta y$, respectively. The relative uncertainties of the data are then expressed as e.g. $\pm 100(\delta x/x)\%$. The rules can be summarized as follows:

Rule 1: The absolute uncertainty of $x+y$ or $x-y$ is obtained by adding the absolute uncertainties δx and δy in quadrature:

$$\delta(x-y) = \delta(x+y) = [(\delta x)^2 + (\delta y)^2]^{1/2} \quad (5)$$

Rule 2: The fractional uncertainty of x times y or x/y is obtained by adding the relative uncertainties δx and δy in quadrature.

$$\delta(xy)/xy = \delta(x/y)/(x/y) = [(\delta x/x)^2 + (\delta y/y)^2]^{1/2} \quad (6)$$

Rule 3: The relative uncertainty on x to a power n is just n times the relative uncertainty of x :

$$\delta(x^n)/(x^n) = n(\delta x/x) \quad (7)$$

In fact, this is true for any power of x , not just positive integer powers. Rules 1 and 2 depend on the assumption that the uncertainties on x and y have nothing to do with each other and may cancel each other out partially. This is clearly not

true if we multiply a number by itself, which is why Rule 3 cannot be derived from Rule 2.

Rule 4: The absolute uncertainty on an arbitrary function of x is obtained by taking the derivative of that function, evaluating it at the value of the measured data, taking the absolute value, and multiplying the result by the absolute uncertainty on x . For clarity, we call the value of the measured data x_0 .

$$\delta F(x_0) = |dF/dx|_{x=x_0} \delta x_0 \quad (8)$$

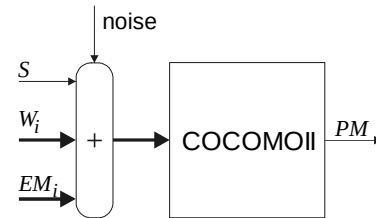


Figure 2. Error propagation

Using the rules described above, error propagation on PM is calculated assuming all the independent variables have the same noise and distribution. After applying the rules (5-8) to the COCOMO II equation (1), the following formula is obtained for determining the relative uncertainty on PM

$$\delta[PM]/PM = (L/1000) \times [17 + \sum W_i^2 \times (\log S)^2 + (\sum W_i)^2]^{1/2}, \quad (9)$$

where the new variable L represents the level of uncertainty of the inputs to the model.

With equation (9), the dependent relative error can be determined when the input variables are added with increasing level of noise by changing L . Based on the first project data, with the standard deviations of all input variables set to $L/1000$ % of the actual value, the standard deviations (SD) of the dependent variable PM are listed in Table 6.

It can be seen that the error propagation is quite notable: when the input error for all independent variables are only 0.1% (10/1000 for $L=10$) of each of them, the error propagated on the dependent variable PM has reached to 7.22%. Therefore, for the preset tolerable error threshold equal to 5%, the input error should be confined within 0.1%.

The resulting values of tolerable input errors for all 15 projects are summarized in Table 7.

Table 6. Standard deviations of *PM* for increasing levels of uncertainty *L* (Project 1)

L	1	2	3	4	5	6	7	8	9	10
SD	0.0722	0.1443	0.2165	0.2886	0.3608	0.4329	0.5051	0.5772	0.6494	0.7215

Table 7. Tolerable input error (TIE) obtained by error propagation

P	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
TIE	0.1%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%

In order to verify the result obtained by error propagation, another Monte Carlo simulation has been designed. Each variable is disturbed by noise with standard deviation starting from 1/1000 of its actual data and increasing by 1/1000 of the data in each step

Table 8: Tolerable input error obtained by error propagation using Monte Carlo Simulation

Project number	Normal Distribution	Normal + Triangular distribution	Normal + Beta distribution
1	0.30%	1.3%	4.2%
2	0.30%	1.4%	4.2%
3	0.30%	1.4%	4.2%
4	0.30%	1.4%	4.2%
5	0.30%	1.3%	4.2%
6	0.30%	1.3%	4.2%
7	0.30%	1.3%	4.2%
8	0.30%	1.3%	4.2%
9	0.30%	1.4%	4.2%
10	0.30%	1.4%	4.2%
11	0.30%	1.4%	4.2%
12	0.30%	1.4%	4.2%
13	0.30%	1.4%	4.2%
14	0.30%	1.4%	4.2%
15	0.30%	1.4%	4.2%

until the upper boundary is reached. The upper boundary of the standard deviation is subject to change with respect to different variables, because they have different defined ranges. The rest of the simulation is analogous to that described in Section 3.2. The tolerable input errors obtained by the Monte Carlo simulation for the threshold of 5% of the resulting effort estimation are listed in Table 8.

The results indicate that the tolerable input errors obtained with normal distribution based Monte Carlo simulation have the smallest values, and that these values are comparable to the results obtained analytically. The tolerable input errors obtained with triangular and Beta distributions have much larger values caused by different properties of such distributions. Though these distributions seem to fit better the nature of effort multipliers and exponent scale factors, the results obtained do not correspond to the analytical solution and are not considered valid. In addition, the size *S* is the most important input variable. The tolerable input errors remain quite stable across different project data for the same distribution type, which means that the 15 data points have been collected in a consistent way.

4. Conclusions

Results obtained for single input noise cases by both mathematical analysis and Monte Carlo simulation consistently indicate that the most critical input variable is size *S*, closely followed by effort multipliers EM_i . The impact of error in determining exponent scale factors W_i is relatively small. This finding can be used to direct management in collecting and estimating these values: Size and effort multipliers are very important and adequate time and resources should be devoted to their accurate evaluation. The scale factors are much less important and could be neglected (set to their nominal values) if necessary.

The results obtained through error propagation are alarming showing very high sensitivity of the model when cumulative errors are considered. In such cases, a relatively small error in determining projected size of the system

could lead to a large error of the estimated value of effort.

To cope with the problems of uncertainty in software cost estimation, authors have proposed to use techniques such as fuzzy sets to represent variables entering the estimation process and relations between them [19]. Rather than using a single number, the software size can be regarded as a fuzzy set yielding the cost estimate also in form of a fuzzy set. Fuzzy sets help articulate the estimates and their essence and generate a feedback as to the resulting uncertainty of the results. The manager is no longer left with a single number estimate that could be highly misleading in many cases and lead to the belief as to the relevance of the obtained results.

5. Acknowledgements

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6. References

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